

# Claude Agent Trader Technology Walkthrough

A detailed PDF guide to the AgentSDK-centered, AI-native architecture built for the separate technology prize. The focus is technical merit: agent orchestration, read-only tools, Anthropic critique, guardrails, traceability, and demo reproducibility.

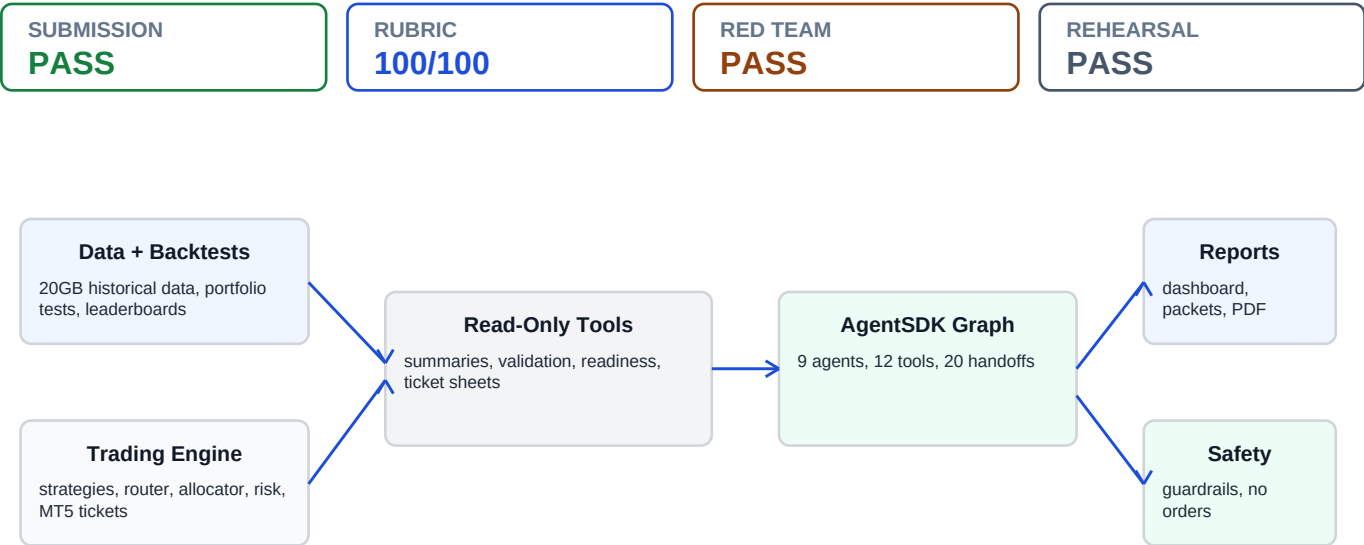


Figure 1. Technology stack overview: trading research remains deterministic; the AgentSDK layer governs evidence, critique, safety, and presentation.

- Built for the separate technology prize, not only trading P&L.
- Default demo path is offline and credit-safe.
- Agent tools are read-only; no MT5 order placement is exposed to models.
- Final bundle generated: 2026-06-27T10:13:42.763095+00:00

# 1. Executive Snapshot

Claude Agent Trader has been wrapped as a technology-prize submission around a safe agentic control plane. The trading engine still supplies research artifacts, backtests, risk outputs, dashboards, and MT5 ticket sheets. The new technology layer makes those artifacts inspectable by specialist agents and proves the boundary between AI reasoning and trading authority.

| Component         | Status | Meaning                                     |
|-------------------|--------|---------------------------------------------|
| Submission bundle | PASS   | Final artifact manifest and command index.  |
| Rubric            | PASS   | Score 100/100 across technology-prize axes. |
| Red-team report   | PASS   | 8 skeptical judge challenges.               |
| Demo rehearsal    | PASS   | 8 offline demo steps.                       |
| Guardrails        | PASS   | 8 executable safety checks.                 |
| Topology          | PASS   | 9 agents and 12 tools.                      |
| Trace replay      | PASS   | 36 span-like events.                        |

## What the judges should understand in one minute

- The system is no longer just a collection of scripts. It has a multi-agent control plane with explicit roles, handoffs, tools, and guardrails.
- OpenAI AgentSDK is the orchestration center: agents inspect evidence through typed, read-only function tools.
- Anthropic credits have a defined role: an independent critic reviews risk, overfitting, and presentation quality without trading authority.
- The safety model is executable. The project can prove that AI tools cannot place MT5 orders and cannot silently spend model credits.
- The demo is reproducible: `tech-prize-rehearse` and `tech-prize-submit` regenerate the proof pack offline.

## 2. System Architecture

The project is organized as a layered system. Lower layers do deterministic trading work: ingest data, run strategies, route signals through risk, backtest portfolios, and prepare MT5 ticket artifacts. The technology-prize layer sits above that engine. It does not replace the trading logic; it supervises and explains it.

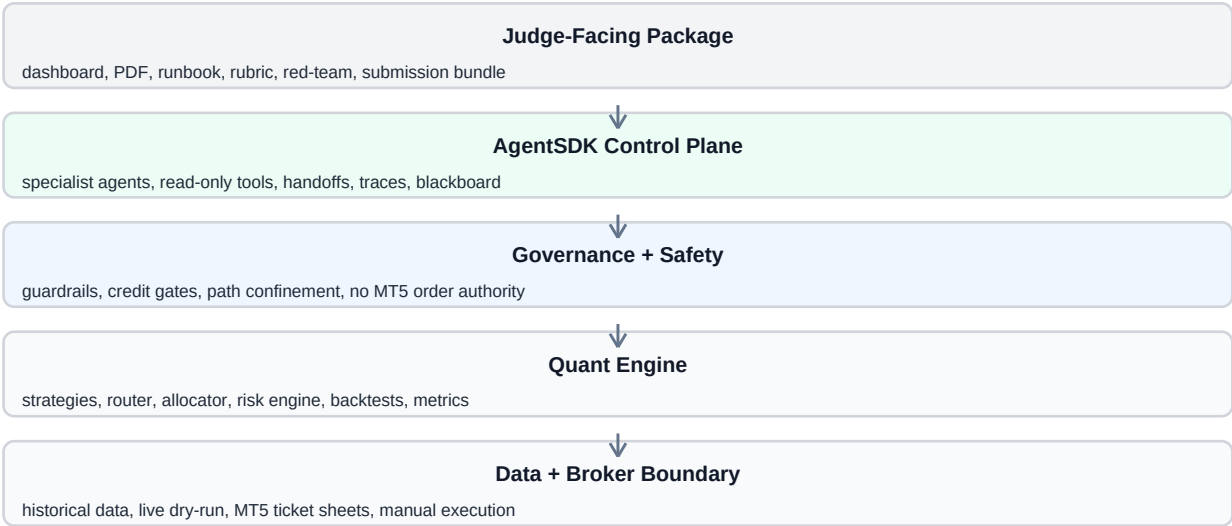


Figure 2. Layered architecture: deterministic quant engine below, agentic evidence and governance layer above.

### Why this matters

A judge can inspect how the system is built without trusting a live model response. Every layer produces artifacts, and every technology-prize claim points back to files, hashes, reports, or executable checks.

### 3. AgentSDK Control Plane

The control plane defines 9 specialist agents, 12 read-only tool specs, and 20 validated handoffs. The Chief Trading Agent coordinates specialist roles instead of letting one prompt do everything.

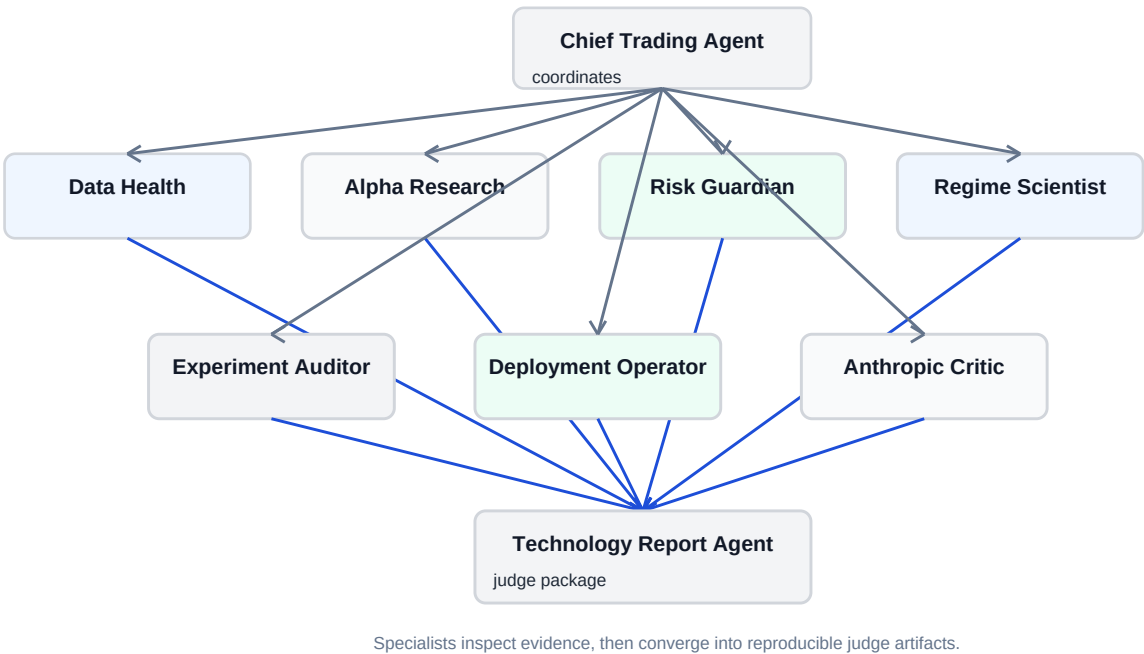


Figure 3. Agent graph: the Chief Trading Agent delegates to specialist roles and all roads lead to judge-facing reporting.

| Agent                     | Technology role                                                               |
|---------------------------|-------------------------------------------------------------------------------|
| Chief Trading Agent       | Coordinates the evidence review and delegates to specialists.                 |
| Data Health Agent         | Checks downloaded data, quote coverage, and market-data quality.              |
| Alpha Research Agent      | Reads research leaderboards, strategy comparisons, and walk-forward evidence. |
| Risk Guardian Agent       | Inspects risk limits, guardrail status, MT5 boundary, and deployment safety.  |
| Regime Scientist Agent    | Explains regime-aware selector and handoff evidence.                          |
| Experiment Auditor Agent  | Checks hashes, provenance, commands, and reproducibility.                     |
| Deployment Operator Agent | Summarizes dashboards, MT5 ticket sheets, and live dry-run readiness.         |
| Anthropic Critic Agent    | Independent critique of overfitting, risk, and narrative quality.             |
| Technology Report Agent   | Packages proof for judges.                                                    |

## 4. Read-Only Function Tool Surface

The AgentSDK bridge exposes evidence tools, not trading controls. This is the main safety and technical-design point: models can inspect project artifacts, but they cannot route orders.

| Tool                                 | Purpose                                                       |
|--------------------------------------|---------------------------------------------------------------|
| summarize_research_artifacts         | Summarize research/deployment files that support claims.      |
| summarize_csv                        | Inspect bounded rows and schema from project-local CSV files. |
| validate_market_data_summary         | Check price/quote data health.                                |
| summarize_experiment_leaderboard     | Read walk-forward and candidate leaderboards.                 |
| build_hackathon_readiness_snapshot   | Summarize competition readiness evidence.                     |
| summarize_mt5_ticket_sheet           | Inspect manual MT5 ticket artifacts without order authority.  |
| summarize_operator_dashboard_sources | Check live operator/dashboard source readiness.               |
| build_technology_prize_judge_packet  | Generate requirement-level evidence matrix.                   |
| run_agent_guardrail_suite            | Run AI/broker safety checks.                                  |
| analyze_agent_topology               | Validate graph, handoffs, tools, and guardrails.              |
| build_judge_demo_runbook             | Build timed demo script and risk answers.                     |
| replay_agent_trace                   | Export local workflow as span-like trace data.                |

- All tools are read-only.
- Paths are confined to the project root.
- No MT5 order-placement tool is registered.
- Online model calls are disabled unless explicitly armed.

# 5. AI-Native Workflow And Traceability

The project treats agent reasoning as inspectable state. The local workflow writes blackboard entries, records handoffs, and exports an offline trace replay. That makes the architecture demonstrable even without spending API credits.

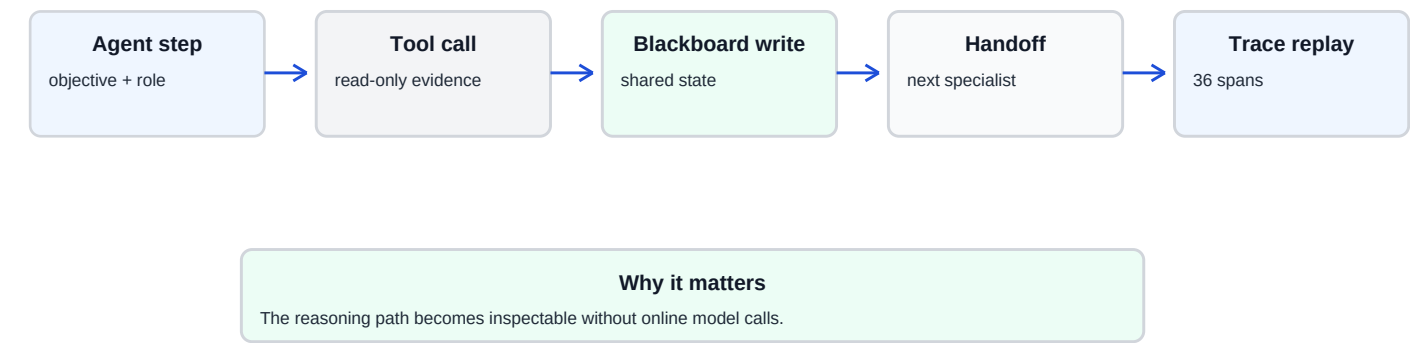


Figure 4. The workflow produces agent steps, tool calls, blackboard writes, and handoffs, then exports trace-like spans.

| Trace kind       | Count |
|------------------|-------|
| agent_step       | 9     |
| blackboard_write | 9     |
| handoff          | 8     |
| tool_call        | 9     |
| trace_root       | 1     |

The most important detail is not the count itself. It is that a judge can see how specialist agents move from evidence to claims through explicit state transitions.

## 6. Safety, MT5 Boundary, And Credit Controls

The technology prize judges may ask whether an AI agent can accidentally trade or spend credits. The answer is no by default. The project makes both boundaries executable: no broker authority is exposed to agents, and online OpenAI/Anthropic calls require explicit flags.

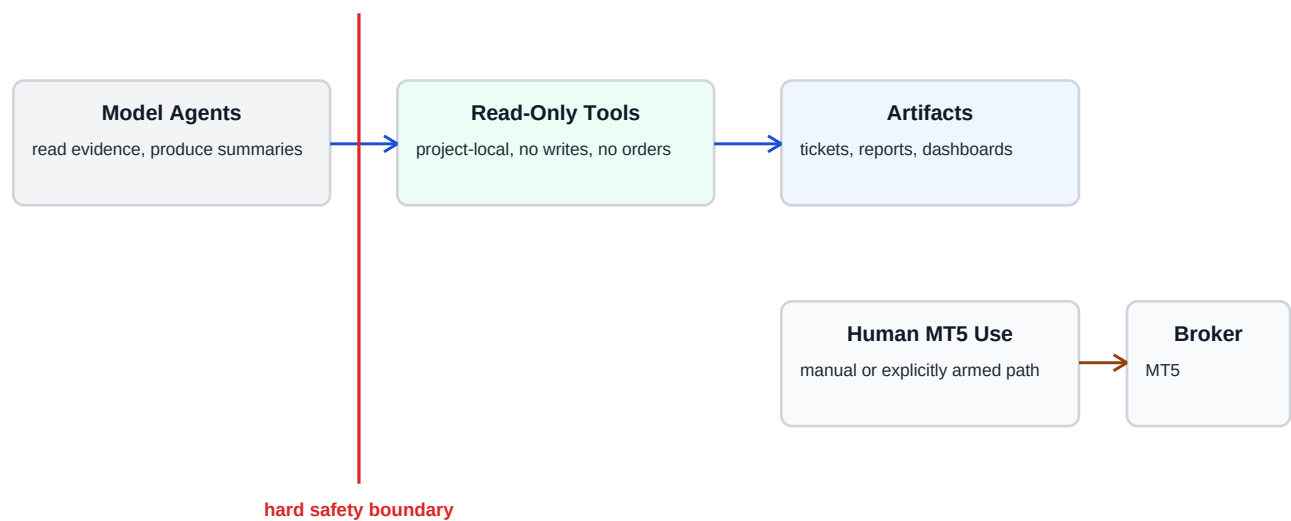


Figure 5. Safety boundary: models inspect artifacts; humans or separately armed systems control live MT5 execution.

| Guardrail                             | Status | Details                                                                       |
|---------------------------------------|--------|-------------------------------------------------------------------------------|
| Read-only AgentSDK tools              | PASS   | 12 tool specs inspected; every tool must be read-only.                        |
| No broker order authority             | PASS   | Workflow blackboard order authority: none exposed to agents                   |
| Project-local path confinement        | PASS   | SDK CSV/artifact tools must reject paths outside the project root.            |
| Online OpenAI calls require arming    | PASS   | AgentSDK Runner calls default to skipped/blocked unless explicitly allowed.   |
| Online Anthropic calls require arming | PASS   | Anthropic critic calls default to skipped/blocked unless explicitly allowed.  |
| Anthropic remains critique-only       | PASS   | Anthropic Critic Agent must not approve trades or receive broker credentials. |
| Evidence before claims                | PASS   | 7/7 required research/deployment artifacts are present.                       |
| Deterministic workflow trace          | PASS   | 9 steps, 9 blackboard writes, 8 handoffs.                                     |

## 7. Anthropic Critic Layer

The Additional Anthropic Credits are not decorative. They have a defined technical role: independent critique. The Anthropic Critic Agent reviews overfitting risk, risk controls, and demo narrative while remaining outside broker authority.



Figure 6. Anthropic is used as a reviewer, not a trader. Its output improves quality assurance and judge readiness.

### Design rules

- Anthropic online calls default to skipped unless `--allow-online-anthropic` is passed.
- The critic cannot approve trades and has no MT5 credentials.
- The critic reviews evidence produced by the deterministic system and AgentSDK control plane.
- This keeps multi-provider AI useful while preserving operational safety.

## 8. Evidence Evaluators

The final system includes evaluators that test the technology story from several angles. This is what makes the submission more than an architecture diagram: the claims are machine-checkable.

| Evaluator    | Status | What it proves                                                                   |
|--------------|--------|----------------------------------------------------------------------------------|
| Topology     | PASS   | Tools are used, handoffs target real agents, and every tool is read-only.        |
| Guardrails   | PASS   | No broker authority, path confinement, model-spend gates, and critic boundaries. |
| Trace replay | PASS   | Agent steps, tool calls, blackboard writes, and handoffs are inspectable.        |
| Rubric       | PASS   | 100/100 score across technology-prize axes.                                      |
| Red team     | PASS   | Skeptical judge objections are answered with evidence checks.                    |
| Rehearsal    | PASS   | Safe demo flow regenerates expected artifacts offline.                           |

### Red-team challenges

| Challenge                                   | Status | Skeptical question                                                                              |
|---------------------------------------------|--------|-------------------------------------------------------------------------------------------------|
| No Hidden Trading Authority                 | PASS   | Can the AI agent secretly place MT5 trades or bypass the risk engine?                           |
| Real AgentSDK Architecture                  | PASS   | Is AgentSDK actually central here, or is this just a normal Python project with an agent label? |
| Anthropic Credits Have A Real Role          | PASS   | Where would Anthropic credits actually improve the system?                                      |
| AI-Native, Not Script-Native                | PASS   | What makes this AI-native instead of many scripts wrapped in a dashboard?                       |
| No Surprise Model Spend                     | PASS   | Could a demo command accidentally spend OpenAI or Anthropic credits?                            |
| Evidence And Provenance Survive Scrutiny    | PASS   | Can the judges verify claims from files instead of trusting the presentation?                   |
| Demo Can Be Reproduced Under Time Pressure  | PASS   | If the live demo gets rushed, is there a reliable proof path?                                   |
| Technology Prize Story Is Separate From PnL | PASS   | If trading P&L is weak, why is this still competitive for the separate technology prize?        |

# 9. Judge Demo Flow

The demo flow is intentionally offline by default. The goal is to avoid surprises during judging while still proving advanced technical architecture.

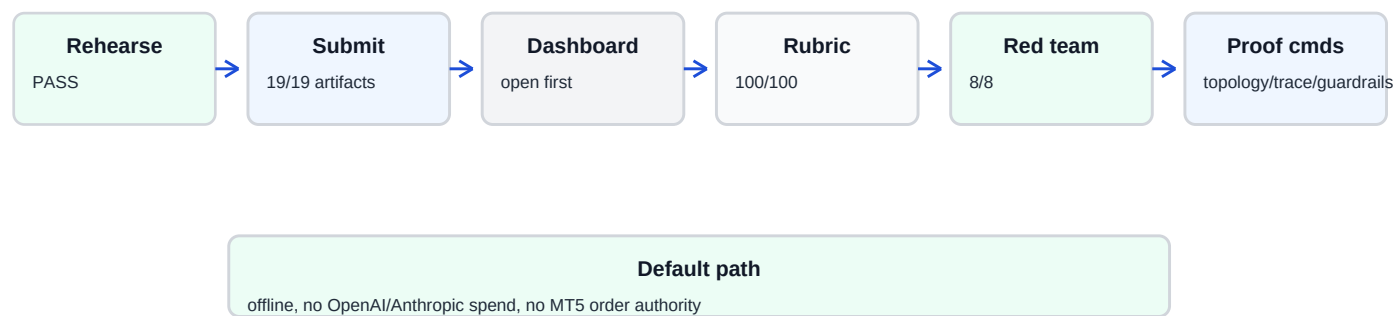


Figure 7. Recommended live flow: rehearse, submit, open dashboard, then run targeted proof commands on request.

| Step | Command                                                     | Purpose                                                                             |
|------|-------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 1    | quanthack tech-prize-pack && quanthack tech-prize-dashboard | Regenerate the main scorecard and browser-first dashboard.                          |
| 2    | quanthack tech-prize-runbook                                | Regenerate the timed judge talk track.                                              |
| 3    | quanthack tech-prize-topology                               | Validate the AgentSDK graph, handoffs, and tool authority.                          |
| 4    | quanthack tech-prize-trace                                  | Export agent steps, tool calls, blackboard writes, and handoffs as spans.           |
| 5    | quanthack tech-prize-guardrails                             | Verify read-only tools, path confinement, credit gates, and MT5 no-order authority. |
| 6    | quanthack tech-prize-judge-packet                           | Verify the prize requirements against source and report evidence.                   |
| 7    | quanthack tech-prize-rubric                                 | Score the project against the technology-prize judging axes.                        |
| 8    | quanthack tech-prize-red-team                               | Answer skeptical judge questions with executable evidence checks.                   |

## 10. Final Artifact Map

The final bundle is the handoff package. It tells the judge what to open first, which commands are safe, and which reports prove each claim.

| Artifact                       | Status | Path                                                 |
|--------------------------------|--------|------------------------------------------------------|
| Open first: dashboard          | OK     | outputs/reports/technology_prize_dashboard.html      |
| Judge demo runbook             | OK     | outputs/reports/technology_prize_demo_runbook.md     |
| Judge demo rehearsal           | OK     | outputs/reports/technology_prize_demo_rehearsal.md   |
| Judge demo rehearsal JSON      | OK     | outputs/reports/technology_prize_demo_rehearsal.json |
| Judge packet                   | OK     | outputs/reports/technology_prize_judge_packet.md     |
| Judge packet JSON              | OK     | outputs/reports/technology_prize_judge_packet.json   |
| Technology prize rubric        | OK     | outputs/reports/technology_prize_rubric.md           |
| Technology prize rubric JSON   | OK     | outputs/reports/technology_prize_rubric.json         |
| Technology prize red team      | OK     | outputs/reports/technology_prize_red_team.md         |
| Technology prize red team JSON | OK     | outputs/reports/technology_prize_red_team.json       |
| Demo pack                      | OK     | outputs/reports/technology_prize_demo_pack.md        |
| Demo pack JSON                 | OK     | outputs/reports/technology_prize_demo_pack.json      |
| AgentSDK topology              | OK     | outputs/reports/technology_prize_topology.md         |
| AgentSDK trace replay          | OK     | outputs/reports/technology_prize_trace_replay.md     |
| Agent guardrails               | OK     | outputs/reports/technology_prize_guardrails.md       |
| Local workflow                 | OK     | outputs/reports/technology_prize_workflow.md         |
| Agent architecture report      | OK     | outputs/reports/technology_prize_agent_report.md     |
| AgentSDK runner report         | OK     | outputs/reports/technology_prize_sdk_runner.md       |
| Anthropic critic report        | OK     | outputs/reports/technology_prize_anthropic_critic.md |

# 11. Appendix: Commands And Talking Points

## Core commands

| Command                                                 | Use                                                   |
|---------------------------------------------------------|-------------------------------------------------------|
| quanthack tech-prize-rehearse                           | Verifies the safe offline demo flow.                  |
| quanthack tech-prize-submit                             | Regenerates the final submission bundle and manifest. |
| open<br>outputs/reports/technology_prize_dashboard.html | Opens the first judge-facing visual artifact.         |
| quanthack tech-prize-rubric                             | Shows the scored prize framing.                       |
| quanthack tech-prize-red-team                           | Answers skeptical judge questions with evidence.      |
| quanthack tech-prize-topology                           | Validates AgentSDK graph integrity.                   |
| quanthack tech-prize-trace                              | Shows trace-like agent spans.                         |
| quanthack tech-prize-guardrails                         | Shows executable AI/broker safety checks.             |

## Closing pitch

- Claude Agent Trader is not just a trading bot. It is an AI-native research and deployment control plane.
- AgentSDK coordinates specialist agents over real project evidence.
- Anthropic credits support independent critique rather than uncontrolled action.
- Guardrails prove that models cannot place trades or spend credits without explicit arming.
- The final demo is reproducible offline and backed by hashed artifacts.